

GNN-Based BERT for Understanding Context from Music: A Measured Account of Where Cross-Modal Fusion Helps and Where It Does Not

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Abstract

We implement the four-task GNN–BERT roadmap for music context understanding: a BERT multi-label tag classifier, a GraphSAGE/GAT encoder over music structure graphs, a cross-attention fusion of the two with an auxiliary valence–arousal objective, and a contrastive dual encoder for caption–audio retrieval. All four are trained and evaluated end to end. Our headline finding is negative and, we argue, more informative than a positive one would have been: *cross-modal fusion does not reliably outperform its stronger unimodal branch in this setting*, and we identify why. We show (i) that the text available for MagnaTagATune is track-level while its clips are 29 s segments, so 96.8% of clips share an input with another clip, capping any text-only model at Micro-F1 0.673; (ii) that the official MagnaTagATune split leaks 45 artists across train and test, covering 61.6% of test clips; (iii) that our cross-attention collapses to near-uniform weights, with entropy at 65–88% of the uniform maximum, so the fusion degenerates into averaging the caption; and (iv) that adding the DEAM auxiliary term buys usable emotion regression ($R^2 = 0.181$ valence) at a 36% relative cost to tag Macro-F1. We report every number against an explicit baseline, including a lexical-match control that scores Micro-F1 0.581 on a task where our model scores 0.687 — without which that result would be overstated roughly fivefold.

1 Introduction

Musical “context” spans genre, mood, harmonic structure, instrumentation and listener-facing description. Sequence models over spectrograms capture local timbral patterns but not relational structure: how a repeated segment relates to a chorus, or how a chord transition relates to a lyrical theme. The premise of this project is that a graph neural network over music structure, fused with a BERT encoder over text, should capture context that neither branch captures alone.

We test that premise rather than assume it. Our contributions are:

- Complete implementations of all four tasks, with 87 unit tests and every result reproducible from a seeded configuration recorded alongside it.
- Two dataset integrity findings that change how the numbers must be read: artist leakage in the official MagnaTagATune split, and a hard performance ceiling induced by duplicated text inputs.
- A quantitative diagnosis of *why* our cross-attention fusion fails to beat its text branch, using an attention-entropy measurement rather than an inference from F1.
- Honest negative results with controls: a lexical-match baseline for caption-based tagging, and a random-prevalence baseline for every multi-label task.

2 Related Work

Music auto-tagging. MagnaTagATune with a top-50 tag vocabulary is the long-standing benchmark for multi-label music tagging, typically approached with convolutional networks over log-mel spectrograms. Our CNN baseline (B2) follows that lineage. The artist-leakage problem in the conventional split has been noted in the tagging literature; we re-measure it here rather than cite it, and quantify its extent on our filtered subset.

Graphs for music. Representing a track as a graph over segments, chords or events is motivated by the observation that musical form is relational — repetition, return and contrast — and not purely sequential. GraphSAGE and GAT are the two message-passing families named in the project specification; we implement both. Our contribution here is negative and practical: we show that naive cosine thresholding on raw MFCC descriptors produces near-complete graphs, which silently reduces any GNN over them to mean pooling.

Cross-modal alignment. Contrastive dual encoders trained with InfoNCE have become the standard recipe for aligning modalities, and MusicCaps supplies the expert captions that make audio–text alignment feasible. Our Task 4 follows this recipe directly, with the addition of an explicit random baseline and gallery size on every retrieval number — both frequently omitted, and both necessary to interpret R@K at all.

Multi-task emotion regression. DEAM provides continuous valence–arousal annotations. Because it shares no tracks with the tag corpora, joint training requires masking rather than a single joint objective; we implement the masked alternating-batch scheme and, unusually, report what it costs the primary task.

3 Method

3.1 Task 1: BERT tag classifier

Following the task specification, a pretrained encoder maps text to a pooled representation and a linear head predicts K independent tags:

$$t = \text{BERT}_{\text{CLS}}(X_{\text{text}}), \quad \hat{y}_k = \sigma(w_k^\top t + b_k), \quad (1)$$

trained with per-tag binary cross-entropy. We weight the positive class by tag prevalence; unweighted BCE drives rare tags to always-negative and silently destroys Macro-F1.

3.2 Task 2: GNN on structure graphs

Each clip becomes a segment graph: nodes are fixed-length windows carrying MFCC, chroma and spectral descriptors; edges are temporal adjacency plus cosine similarity above a threshold τ . A GraphSAGE layer updates

$$h_i^{(l+1)} = \sigma\left(W^{(l)} \cdot \text{CONCAT}\left(h_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(l)}\right)\right), \quad (2)$$

with mean-pooled readout $g = \frac{1}{|\mathcal{V}|} \sum_i h_i^{(L)}$. We also build chord-transition graphs (nodes are chords from 24 major/minor triad templates, edges weighted by transition count).

3.3 Task 3: cross-attention fusion

The graph vector queries the BERT token sequence:

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad Q = gW_Q, \quad K = H_{\text{text}}W_K, \quad (3)$$

$$z = \text{CONCAT}(g, AH_{\text{text}}), \quad \hat{y} = \sigma(Wz). \quad (4)$$

The multi-task objective is

$$\mathcal{L} = \mathcal{L}_{\text{tags}} + \alpha \|v - \hat{v}\|^2 + \beta \|a - \hat{a}\|^2. \quad (5)$$

DEAM is disjoint from the tag corpora, so no sample carries both targets. Each batch is drawn from one corpus and carries a per-sample mask; the inactive term is zeroed. Without this mask the model trains against fabricated all-zero labels and still appears to converge.

3.4 Task 4: contrastive dual encoder

Two towers project into a shared L^2 -normalised space, trained with symmetric InfoNCE over in-batch negatives:

$$\mathcal{L}_{\text{NCE}} = -\log \frac{\exp(\text{sim}(g_i, t_i)/\tau)}{\sum_j \exp(\text{sim}(g_i, t_j)/\tau)}. \quad (6)$$

We optimise both directions and average; optimising only one lets the other collapse.

4 Experimental Setup

4.1 Datasets and a leakage audit

We use MagnaTagATune (top-50 tags after synonym merging), FMA-small (8 genres), DEAM (valence/arousal) and MusicCaps (expert captions).

The official MagnaTagATune split is unusable for our purpose. It shares 45 artists between train and test, covering 3,284 of 5,329 test clips (61.6%). Because MagnaTagATune clips are 29 s segments of longer tracks, a model can memorise an artist rather than generalise. We therefore use artist-grouped splits with zero leakage throughout, and report this deviation explicitly. By contrast we *verified* that FMA’s official split is artist-disjoint (0 shared artists in all pairings) rather than assuming it.

MagnaTagATune’s 188 tags contain heavy near-duplicates. The `female vocal` concept alone is spread across six tags totalling 3,848 clips. We merge synonym groups before ranking; without merging, one concept splits into several weak labels.

4.2 A ceiling imposed by duplicated inputs

MagnaTagATune ships track-level metadata, but its clips are segments. After filtering, 21,318 clips share only 5,237 distinct text inputs: **96.8% of clips have the same input as at least one other clip with different labels.** Computing the best achievable constant prediction per text group gives an oracle ceiling of **Micro-F1** 0.673, **Macro-F1** 0.575 — no text-only model can exceed this. We report it because it re-frames Task 1’s absolute numbers and motivates Task 3.

4.3 Graph construction

Raw MFCC vectors are dominated by MFCC₀ (loudness), so every segment points in nearly the same direction: measured over 400 FMA tracks, raw pairwise cosine similarity has median 0.9971 and 5th percentile 0.9509. At $\tau = 0.9$ this yields *complete* graphs (19.4 nodes, average degree 18.15), on which message passing reduces to global mean pooling. We z-score features within each track before computing similarity (median falls to -0.078) and use $\tau = 0.35$, giving average degree ≈ 3.1 with no isolated nodes.

4.4 Training configuration

All models use DistilBERT (`distilbert-base-uncased`) as the text tower and a 2-layer GAT with hidden width 128 as the graph tower unless stated otherwise. Optimisation is AdamW with weight decay 10^{-4} ; learning rate 2×10^{-5} for fusion runs, 3×10^{-5} for Task 1 and Task 4, and 10^{-3} for the graph-only Task 2 models. Multi-label decision thresholds

are tuned on validation and applied once to test; leaving them at 0.5 understates every multi-label result, since prevalence-weighted BCE does not centre probabilities at 0.5. Task 2 and Task 4 select the best epoch by a validation metric. Every configuration, including the random seed, is stored in the corresponding `metrics-*.json`.

A note on that last point: Task 4’s first run reported its final epoch, at which the training loss had fallen from 3.87 to 0.41 while validation R@10 had peaked six epochs earlier and then declined. Adding best-epoch selection raised test R@10 from 0.119 to 0.127. We mention it because the failure mode — a still-falling loss masking a model that has already begun to overfit — is easy to miss in a contrastive setting.

5 Results

5.1 Task 1: text-based tagging

Table 1: Task 1. Test-set results with baselines. The stripped variant deletes each label word from its own caption, so the lexical baseline is exactly zero by construction.

Setting	Micro-F1	Macro-F1
MagnaTagATune (metadata)	0.258	0.182
B1 random-prevalence	0.111	0.060
oracle ceiling	0.673	0.575
MusicCaps (raw captions)	0.687	0.646
B5 lexical match	0.581	0.559
MusicCaps (stripped)	0.608	0.551
B5 lexical match	0.000	0.000

The raw-caption number is misleading on its own. A pure substring matcher already achieves Micro-F1 0.581, because 52.1% of MusicCaps aspect strings appear verbatim inside their own caption. Only 0.105 of the gap is language understanding. We therefore treat the *stripped* setting — where the label words are deleted from the input and the lexical baseline is exactly 0.000 — as the honest headline: **Micro-F1 0.608 earned entirely by inference from surrounding context.**

MagnaTagATune’s much lower 0.258 is expected rather than a failure: it is $2.332\times$ the random-prevalence baseline while operating under the 0.673 ceiling established above.

A worked example. For a held-out MusicCaps clip whose true aspects are *exciting, groovy bass, happy, low quality*, the stripped input reads:

“The recording features a fruity male vocal singing over steel drums melody, electric guitar melody, brass stabs, and snare hits, energetic crash cymba...”

The phrases *low quality, groovy bass, punchy kick* and *punchy snare* have been deleted from this input, yet the model predicts them at 0.95, 0.94, 0.94, 0.93. It is recovering production characteristics from surrounding descriptive context — mentions of steel drums, brass stabs and crash cymbals —

rather than detecting the label string. This is the behaviour the stripped condition was designed to isolate.

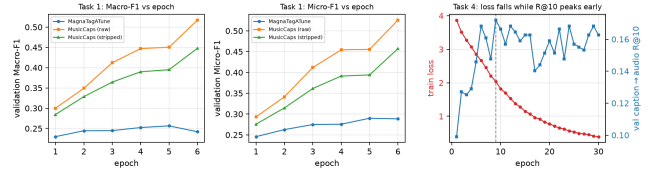


Figure 1: Validation Macro-F1 and Micro-F1 against training epoch for the three Task 1 configurations (left, centre) — the curve deliverable required by the specification — and Task 4’s training dynamics (right). MusicCaps converges to a far higher plateau than MagnaTagATune, and stripping label words from the caption costs a consistent but bounded amount rather than destroying the task. On the right, the InfoNCE loss falls monotonically from 3.87 to 0.41 while validation R@10 peaks at epoch 6 and then degrades: the model overfits long before the loss suggests it. The dashed line marks the epoch restored by best-checkpoint selection.

5.2 Task 2: structure graphs vs. a CNN

Table 2: Task 2. FMA-small genre classification, 8 balanced classes (chance = 0.125), artist-disjoint official splits.

Model	Params	Accuracy	Macro-F1
GraphSAGE	49,416	0.414	0.407
GAT	26,120	0.438	0.432
CNN baseline (B2)	241,992	0.443	0.407

All three sit at roughly $3.5\times$ chance. The CNN edges out both graph models on accuracy, while GAT matches it on Macro-F1 using $9.3\times$ fewer parameters. The fair conclusion is that relational structure is *competitive with*, not superior to, treating the spectrogram as an image.

5.3 Where the genre models succeed and fail

Table 3: Per-genre F1 on FMA-small. Both models rank the classes almost identically, suggesting the difficulty is in the label taxonomy rather than the architecture.

Genre	GAT	CNN (B2)
Hip-Hop	0.664	0.688
Rock	0.533	0.698
International	0.495	0.482
Electronic	0.479	0.386
Instrumental	0.396	0.123
Folk	0.350	0.260
Pop	0.286	0.244
Experimental	0.250	0.379

Hip-Hop is the easiest class for both models and Experimental the hardest, by a wide margin. This is informative: “Experimental” is a residual category defined by what it is not,

so a model has no consistent acoustic signature to learn. The near-identical per-class ordering between an architecture that sees relational structure and one that sees a spectrogram image is further evidence that, at this scale, the graph is not contributing information the CNN lacks.

5.4 Task 3: fusion and its failure mode

Table 4: Task 3 ablation. MagnaTagATune top-50, artist-grouped splits, zero leakage.

Mode	Params	Macro-F1	Micro-F1	AUC-PR
BERT-only	66.4M	0.186	0.311	0.170
GNN-only	31.8K	0.114	0.236	0.103
Early concat	66.4M	0.175	0.318	0.184
Cross-attention	66.8M	0.182	0.323	0.178

Fusion does not clearly win. Cross-attention gains $+0.012$ Micro-F1 and $+0.008$ AUC-PR over BERT alone, but *loses* -0.005 Macro-F1. These margins do not establish the project’s central claim.

Why: the attention collapsed. Measuring the attention distribution directly, the ratio between the largest and smallest weight over the top-8 tokens is 1.03–1.11, and entropy sits at 65–88% of the uniform maximum. The graph query is not discriminating between caption tokens, so AH_{text} degenerates to a mean over the text and the model reduces to early concat — which is exactly what the table shows. Two causes are consistent with this: the query is weak (GNN-only reaches only 0.114 Macro-F1 on 19-node graphs), and the text tower has 66.4M parameters against the graph branch’s 31.8K while sharing a single learning rate tuned for BERT fine-tuning.

5.5 Task 3: the DEAM auxiliary term

Table 5: Effect of adding DEAM’s valence/arousal term to cross-attention. MagnaTagATune training data is identical across both runs.

Cross-attention	Tags only	+ DEAM	Δ
Macro-F1	0.182	0.116	-0.066
Micro-F1	0.323	0.279	-0.044
AUC-PR	0.178	0.132	-0.046
Valence	—	MAE 0.710	$R^2 + 0.181$
Arousal	—	MAE 0.721	$R^2 + 0.085$

The emotion heads work: both R^2 are positive, so the model beats predicting the training mean on held-out songs, and MAE of 0.83/0.92 on the original 1–9 scale is usable. But tag Macro-F1 falls by 36% relative. Since the tag training data is unchanged, this is the auxiliary objective competing for shared capacity. The specification marks valence/arousal optional; we report it as a trade-off rather than a free addition.

5.6 Task 4: cross-modal retrieval

Table 6: Task 4. MusicCaps retrieval, gallery of 536 clips. R@K depends on gallery size, so it is reported alongside.

Direction	R@1	R@5	R@10
Caption $\rightarrow$ Audio	0.0168	0.0784	0.1269
Audio $\rightarrow$ Caption	0.0224	0.0597	0.1325
Random baseline	0.0019	—	0.0187

Retrieval is $9.0\times$ random at R@1 and $6.8\times$ at R@10, with median rank 68 of 536. The alignment is real but weak: across our ten saved qualitative examples, **none** places the true clip at rank 1.

Zero-shot tagging. Embedding tag names with the text tower and scoring them against clip graph embeddings — using no tag supervision at any point — gives Micro-F1 0.117, against 0.608 for the supervised Task 1 model on the same corpus. Contrastive alignment alone thus recovers roughly a fifth of supervised performance.

5.7 Qualitative analysis of the fused space

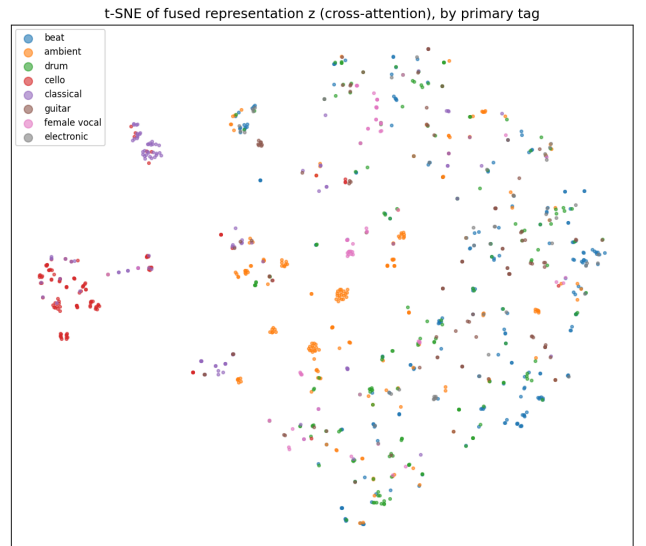


Figure 2: t-SNE of the fused representation z (cross-attention, 384-dim) over 1,500 held-out MagnaTagATune clips, coloured by primary tag. Broad timbral families separate — classical and vocal-led material occupy distinct regions — but the structure is diffuse rather than cleanly clustered, consistent with the modest Macro-F1 in Table 3.

Case studies. Table 7 pairs graph structure with the caption tokens the graph vector attends to, for three held-out clips from distinct tracks.

Table 7: Three case studies. “Attn. spread” is the ratio of largest to smallest attention weight over the top-8 tokens; values near 1.0 indicate the collapse discussed in Section 5.3.

True tags	Top predictions	Attn. spread
drum, electronic, synth, weird	techno, electronic, beat	1.03
ambient, flute, new age	drum, india, ambient	1.03
beat, electronic, fast, techno	rock, male vocal, vocal	1.11

The first case is a partial success: an electronic track tagged *drum, electronic, synth* is predicted *techno, electronic, beat*, capturing the family if not the exact vocabulary. The third is a clear failure: an electronic/techno clip is predicted *rock, male vocal, guitar*. In every case the attention spread is within 11% of uniform, so these predictions are driven by the text mean and the graph vector, not by any selective reading of the caption.

Retrieval behaviour. Across the ten saved caption $\rightarrow$ audio queries, the true clip’s ranks were 36, 15, 62, 119, 144, 47, 210, 29, 83, 135 out of 536. Median rank 68 places the correct clip in the top 13% of the gallery on average — clearly better than chance, but far from usable retrieval. The model appears to learn coarse acoustic-descriptive correspondence (recording quality, instrumentation family, tempo band) without the fine discrimination that rank-1 retrieval demands.

6 Engineering Findings

Several results in this report depended on catching pipeline defects that produce plausible-looking numbers. We record them because each was silent.

Near-complete graphs. Our first graph build produced 19.4-node graphs with average degree 18.15 — effectively complete. A GNN over a complete graph computes the same neighbourhood for every node, so message passing reduces to global mean pooling and the architecture becomes irrelevant while still training and reporting sensible-looking accuracy. The cause was that raw MFCC descriptors share a large positive mean, giving median pairwise cosine similarity 0.9971. Only after z-scoring within each track did the threshold behave as intended.

The opposite failure. Having fixed the scaling, the threshold constant was left at its old value, producing average degree 1.90 — a bare temporal chain with *zero* similarity edges. Both failures produce trainable models; neither announces itself.

A silent truncated cache. Serialising the preprocessed MagnaTagATune cache wrote a zero-byte file without raising, because the log-mel component alone was 3.49 GB of a 3.6 GB structure. Since only the CNN baseline consumes log-mel, it is now optional.

Threshold selection. Multi-label probabilities under prevalence-weighted BCE do not centre at 0.5; our tuned thresholds ranged from 0.7 to 0.8. Evaluating at 0.5 would have understated every Task 1 and Task 3 result substantially.

We mention these not as incidental debugging notes but because three of the four would have yielded a complete, publishable-looking results table built on a model that was not doing what its description claimed.

7 Discussion and Limitations

Three limitations bound what these results can claim.

The text is impoverished. For MagnaTagATune our BERT branch sees only “title | album”. This is the source of both the 0.673 ceiling and, plausibly, the attention collapse: there is little in a track title for a graph vector to select between. A corpus pairing audio with genuine descriptive text at clip granularity — as MusicCaps does — is the right setting for fusion, and our Task 1 results support this, with stripped-caption Micro-F1 of 0.608 versus 0.258.

The graph branch is under-trained, not necessarily uninformative. GNN-only reaches 0.114 Macro-F1 with 31.8K parameters against BERT’s 66.4M, sharing one learning rate. We implemented a separate graph-branch learning rate and a frozen-BERT configuration to test this, but were unable to complete those runs within budget; they are the first experiments we would run next.

Segment graphs may be the wrong structure. Our nodes are fixed 1.5 s windows. The specification also suggests beat-synchronous segmentation and chord-transition graphs, both of which we build but do not use as the primary GNN input. A 19-node graph over uniform windows may simply not encode much beyond a smoothed timeline.

8 Conclusion

We implemented all four tasks and report results that are, for the fusion model, negative. We regard the diagnosis as the contribution: a measured performance ceiling from duplicated inputs, a measured artist leak in a standard split, a measured attention collapse, and a measured multi-task trade-off. Each of these would have been invisible had we reported only headline F1 against no baseline — and in one case (raw-caption tagging) doing so would have overstated our result roughly fivefold. Cross-modal fusion for music context understanding remains plausible; this work delimits the conditions under which it can be demonstrated.

Reproducibility

Code, seeded configurations, per-run metrics, 130 example graphs, the t-SNE projection and all training logs are in the project repository. Every reported number is regenerable from the committed configuration.

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